

EE 592P: Selected Topics in IoT

Course Code	EE 592P
Course Name	Selected Topics in IoT
L-T-P-C	0-1-2-2
Prerequisites	IC 161 – Applied Electronics or equivalent; IC 250 – Programming and Data Structure Practicum or equivalent; Consent of instructor required
Intended for	B.Tech. (EE & CSE) IV year/ PG
Distribution	Elective for B.Tech. IV year, MS, MTech, PhD
Semester	Even/Odd
Approval	recommended by the 20th BoA (Dec 2017)

Preamble

Internet-of-Things is a rapidly expanding domain with applications in both the national and international context. This practicum course is intended to give the students a hands-on experience of end-to-end design and implementation of IoT-based systems. It covers issues related to data acquisition and practical deployment of IoT-based systems, connecting the sensors to the cloud, and analytics on the cloud. The course will involve 12 weekly laboratory sessions and one final project. By the end of the course students will be proficient in interfacing sensors with common microcontrollers such as Arduino and Raspberry Pi, enabling device-to-device communications, real time data acquisition and hardware control through sensor networks, and connecting the physical systems to a cloud computing platform.

Course Modules

- **Module 1: Hardware components of IoT systems:** Introduction to microcontrollers including PIC, Arduino, Raspberry Pi; interfacing sensors and actuators with microcontrollers, building breadboard circuits, using IoT CAD tools; performance characteristics, drift, measurement techniques, packaging; edge analytics and sensor control; (6 tutorial hours + 12 lab hours)
- **Module 2: Networking of devices:** Device-to-device communication; networking protocols and architectures; scheduling and routing; wired and wireless (Bluetooth, Zigbee, NFC) sensor networks; communication technologies like LoRA, SigFox, Cellular IoT; transferring data to the cloud; (5 tutorial hours + 10 lab hours)
- **Module 3: Final Project:** Creation of application specific (e.g. agriculture, river monitoring, disaster prediction etc.) IoT systems integrating components from modules 1-2 listed above. (3 tutorial hours + 6 lab hours)

Text Books

1. Donald Norris, *The Internet of Things*, McGraw Hill Education, 2015.
2. Peter Waher, *Learning the Internet of Things*, Packt Publishing Ltd, 2015.

References

1. Upton and Halfacree, *Raspberry Pi User Guide*, Wiley, 2014.
2. Robert Faludi, *Building Wireless Sensor Networks*, O'Reilly Media, 2011.